

Youngjin Yang

Undergraduate Researcher | Life Sciences
yyjyyj1212@naver.com | TOEIC 820 | github.com/yyj1212-art | yyj1212-art.github.io

RESEARCH INTEREST

I am interested in engineering biological systems by combining experimental research with quantitative analysis. My recent work in RNA engineering and circRNA translation has led me to explore how sequence design, mammalian-cell engineering, and data-guided experimentation can improve biological performance.

EDUCATION

Chung-Ang University B.S. in Life Sciences	2026-Present	GPA 4.20 / 4.50
Dongguk University Life Sciences	2022.03-2024.02	GPA 4.09 / 4.50
COMBINED GPA 4.112 / 4.50		

RESEARCH EXPERIENCE

Chung-Ang University - Kim Hakyun Lab

Undergraduate Research Intern | Mar 2026-Dec 2026

- **Focus:** circRNA translation optimization through IRES screening and UTR optimization.
- **Experimental work:** HEK293T cell culture and transient transfection, molecular cloning, and dual-luciferase assays.
- **Analysis:** Quantitative evaluation of IRES candidates and repeatability across experiments.

Dongguk University - Im Gun-il Lab

Undergraduate Research Intern, Regenerative Medicine Research Lab | Jun 2023-Jan 2024

- **Cell-based research:** Regenerative medicine research, spheroid culture, FACS-based evaluation of cellular states, and bone-marrow preparation.
- **In vivo research:** Participated in mouse and rabbit osteoarthritis-model experiments evaluating therapeutic cell- and drug-related effects, including GFP-based observation.

SELECTED RESEARCH PROJECTS

circRNA Translation Optimization | Chung-Ang University | 2026

Used a Firefly/Gaussia dual-luciferase reporter system to compare IRES candidates, with FHB as the WT normalization reference. Follow-up work examined selected IRES and UTR combinations, repeatability, and sequence/structure interactions.

SZ1 IRES Mutation Landscape | Independent Project | 2026

Generated and analyzed 1,950 single-nucleotide substitutions across the 650-nt SZ1 IRES. Used Python and RNAfold to calculate MFE, delta MFE, pairing-state changes, and local structure differences, then connected structural features to nine experimental labels for exploratory analysis. Built a GitHub-hosted interactive landscape to inspect variants and computed features. With only nine experimental labels, the machine-learning analysis remains exploratory and is not a validated prediction model.

IRES Screening & Reproducibility Analysis | 2026

Screened approximately 100 candidate conditions in HEK293T cells. Technical duplicates were measured after 24 hours and normalized to WT (FHB) using Gaussia/Firefly ratios. Repeated screening was used to assess reproducibility and select candidates for follow-up optimization.

TECHNICAL SKILLS

Cell & Molecular Biology

HEK293T cell culture · transient transfection · Lipofectamine 3000 · dual-luciferase assay · FACS · PCR · molecular cloning · plasmid preparation

RNA & Biological Systems

circRNA engineering · IRES screening · UTR optimization · reporter assays · experimental design · reproducibility analysis

Data Analysis & AI-assisted Workflows

Python · quantitative and statistical analysis · RNAfold · feature engineering · AI-assisted coding and debugging · data visualization · interactive mutation-landscape development · GitHub

LEADERSHIP & ACTIVITIES

Student Council: Two years of student council activities; served as a department/student council leader.

External & Campus Projects: Korea Social Contribution Association supporter activities and leadership in curriculum-development and creative-convergence projects.

CAREER DIRECTION

My research goal is to combine AI-assisted experimental design and automated laboratory workflows to improve research efficiency, while using genetic engineering to control mammalian-cell function and enhance productivity. I hope to develop these interests through research on cell engineering and biopharmaceutical production.